EXPERIMENT 3: POWDER METALLURGY

Purpose of Powder Metallurgy: Metal and metal alloys are made a durable body with the help of pressure and temperature without melting their powder. This heat treatment, called sintering, replaces melting and is carried out at a temperature below the melting point of the metal powder used. In the sintering of single component powders, the sintering temperature can be lower than the melting temperature of the material (approx. 80% of the melting temperature). In multicomponent systems, the sintering temperature can be selected from the components just below the melting temperature of the lowest melting temperature. This type of sintering is called solid phase sintering. In addition, sintering temperature in multi-component systems can be taken above the melting temperature of at least one of the components, such sintering processes are also called liquid phase sintering.

Why Powder Metallurgy Manufacturing Method?

The parts manufactured with T / M can be examined in two main groups depending on the preferred reason of this method.

Group 1: Parts of which the T / M method is more economical, although they can be manufactured by other methods.

Group 2: T / M's only alternative parts

Powder Metallurgy Processes

Powder metallurgy technique consists of 3 stages.

1. Powder production,

2. Pressing (powder particles into a single park with various operations),

3. Sintering and final parts production with secondary operations if necessary.

PART 1.1: APPARENT DENSITY (WET DENSITY)

1.1.1 PURPOSE OF EXPERIMENT: Determining the loose, uncompressed density of the powder and determining the conformity to the standards.

1.1.2 TOOL, DEVICE AND MATERIALS USED: Scale, metal powder, standard test apparatus

1.1.3 THEORETICAL KNOWLEDGE AND EXPERIMENT: The bulk density is the loose (uncompressed) density of the powder (g / cm^3). In order to determine this density, the size and shape of the powder is poured from a funnel specified in the standards and is provided to fill the cylindrical container with a free fall. The height between the funnel outlet and the upper surface of the cylindrical container is a constant value. Carefully peel off the excess powder in the cylindrical container which is filled and weigh the powder inside. Since the test

container volume is known (25 cm^3), the bulk density is easily

calculated as (g / cm^3). All conditions related to this test and the conduct of the test are specified in the TS EN ISO 3923-1 standard. In Figure 1, the dimensions of the equipment used for this experiment are given schematically.The apparent density of stainless steel powders produced by water atomization varies between 2.80-3.20 g / cm^3. This value can be up to 5 g / cm^3 for dusts produced by gas atomization. This value varies between 2.80-3.00 g / cm^3 for copper powders and 3.00-3.25 g / cm^3 for bronze powders. The bulk density is a very important factor in filling the mold of the powder during the pressing step. The bulk density is closely related to the shape, size and distribution of the powder grain. As the grain shape moves away from the sphere, the intergranular space rate decreases.

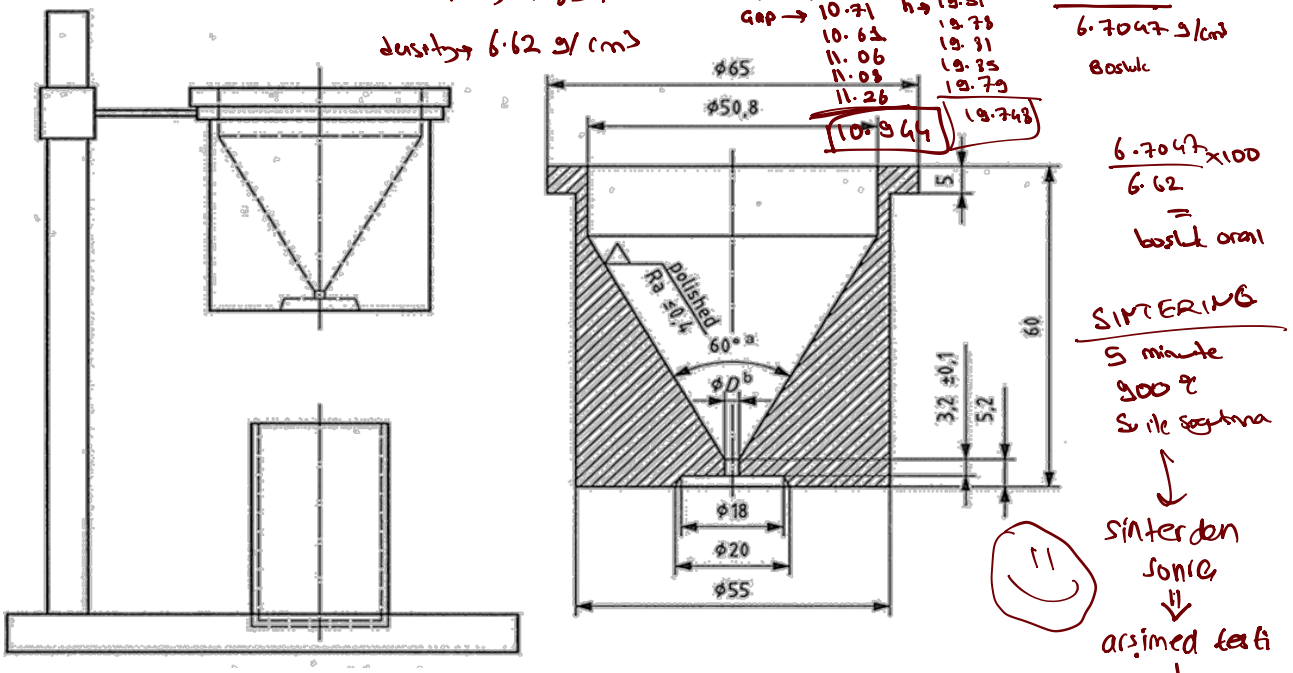


Figure 1.1 Measurement of Masonry Density

PART 1.2: FLUIDITY TEST

1.2.1 PURPOSE OF EXPERIMENT: Determination of the desired flow time from the powders and their compliance with the standards

1.2.2 TOOL, DEVICE AND MATERIALS USED: Scale, metal powder, standard test equipment, stopwatch

1.2.3 THEORETICAL KNOWLEDGE AND EXPERIMENT

The purpose of this test is to determine the fluency values of raw materials from X company.This rate is defined as the time that 50 grams of powder must pass through a 2.54 mm funnel. It has a flow time of 15 seconds for spherical stainless steel. This ratio varies from 25 to 30 seconds for irregularly shaped powders.

Fluency; the size and shape of a given amount of a powder type or mixture thereof is the ability to flow from a predetermined funnel. In these measurements, the flow time of the powder sample of 50 g of powder is generally determined and this value is considered as the flux of the powder. Funnel dimensions and shape with flow hole dimensions (hole diameter and length) are specified in the standards (TS EN ISO 4490). The experiment is usually repeated three times and the mean value is calculated. Its fluency also depends on the powder size, grain size, specific surface size and powder grain shape. Fluency increases as the grain size decreases.

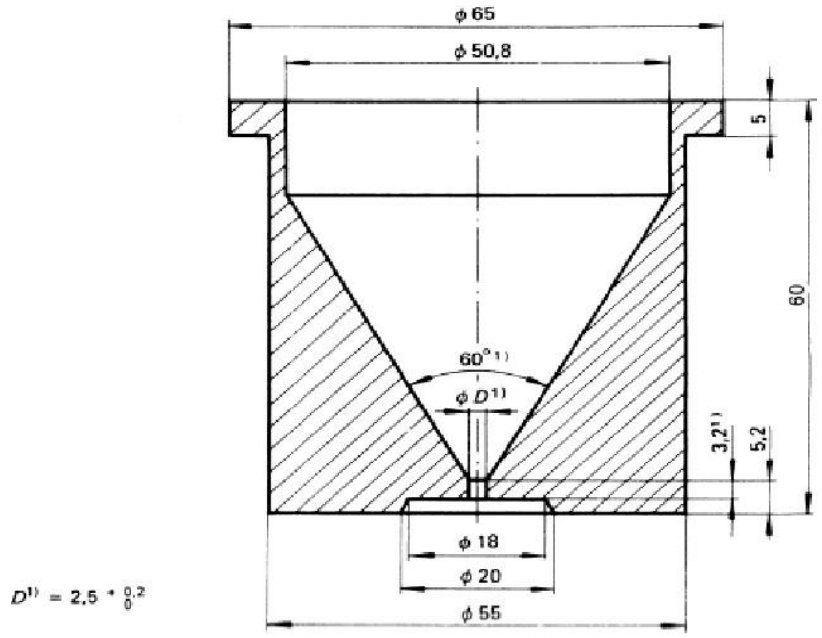


Figure 1.2 An Experimental Funnel for Determining Fluency is Shown Schematically.

PART 1.3: COMPRESSING

1.3.1 PURPOSE OF EXPERIMENT: Determination of pressure-density relationship for different pressures by shaping powders in molds

1.3.2 TOOL, DEVICE AND MATERIALS USED: Metal powder, mold, press

1.3.3 THEORETICAL KNOWLEDGE AND EXPERIMENT

Densification in Powder Metallurgy is the second important process. Pressing is that prepared metal powder mixture is compressed to reach the predetermined density in the appropriate press. It forms the metal powder filled into the press mold into a solid object. In order to achieve a high degree of pressing, a well-formed powder and high pressing pressures are required. The density achieved by pressing affects the mechanical properties of the parts produced by powder metallurgy in the first degree.

Pressing is mostly cold (room temperature), but in special cases it is also hot. By pressing, the metal powder is given a desired level of density and mechanical strength according to the size and shape of the piece to be produced.

There are three basic steps in the mold condensation process:

1. Fill the mold cavity with the specified amount of metal powder,

2. Compression of the powder by means of staples,

3. Removing the shaped part from the mold.

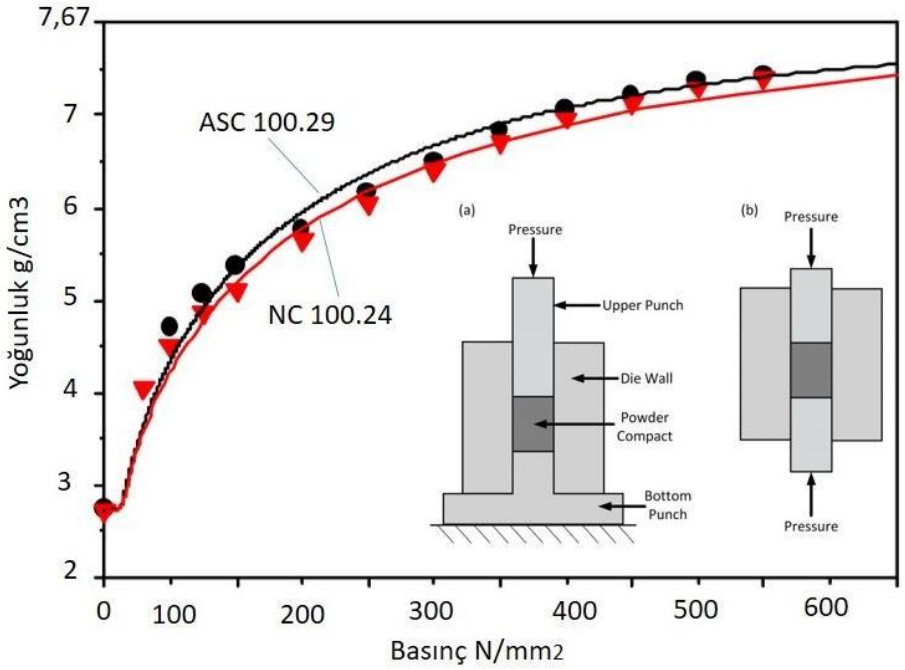


Figure 1.3 Density-pressure relationship for two commercial iron powders

* pressure ↑ , density ↑ , pore ↓

* pore ↑ , density ↓

* pore ↑ , strength ↓

* pore ↑ , lubricant ability ↑

Porosity: Sintered bushing, oil pump rotor, gears, bearing material etc. used in places. (After the production of the parts is completed, the powders in the pore are vacuumed, oil is given under vacuum, air is taken from the pores and oil is pumped into the cavities).

PART 1.4: COMPRESSIBILITY (GREEN DENSITY)

1.4.1 PURPOSE OF EXPERIMENT: Determination of the theoretical density of the powder after pressing.

1.4.2 TOOL, DEVICE AND MATERIALS USED: Scales, metal powder, press, caliper, mold.

1.4.3. THEORETICAL KNOWLEDGE AND EXPERIMENT

This property corresponds to the density obtained when the powder is under pressure. This is the density reached after pressing. Austenitic stainless steel powders have better compressibility than martensite. compressibility; The yield limit is a function of the sample's hardness, the softness of the particles, the void ratio of the sample, the shape and size of the grains.

If the metal powders are formed in the mold, it is provided that the powder free to fill the mold cavity very well and fully (high fluency capability) as close to the theoretical density as when it is pressed into the mold cold (when stapled).

The softer the material, the higher its compressibility. The compressibility is also closely related to the friction between the beads and the molds in the pressing of the powder grains.

The theoretical density of the component: for example, the green density obtained for Fe is 6.65 g/cm^3, Fe: 7.87 g/cm^3: (6.65: 7.87) * 100 = 84.5% = % theoretical density.

- 84,5% Filled

- 15.5% Pore

PART 1.5: SINTERING PROCESSES APPLIED TO METAL POWDERS

1.5.1 PURPOSE OF EXPERIMENT: Effect of sintering process conditions on structure and properties

1.5.2TOOL, DEVICE AND MATERIALS USED: T / M sample, atmosphere controlled oven.

1.5.3 THEORETICAL KNOWLEDGE AND EXPERIMENT

Sintering is the heat treatment applied on the powder without disrupting the shape of material. Prestige shaped powder is not suitable for use in this form. However, the required strength increases with sintering. In this process, diffusion occurs between the powder grains at certain temperature and time, and weak mechanical bonds become strong mechanical bonds.

Sintering temperature and time; A suitable sintering temperature and time should be determined according to properties desired for the parts. The main reason for the low density and strength in the parts is the very low sintering temperature and time. The sintering temperature should be as close as possible to the melting temperature of the powder.

Table 1. Sintering temperature and time of some metals and alloys

Powder materials	Sintering temperaturee (°C)	Sintering Temperature Standby Time (min)
bronzes	760-871−820	10-20
Brasses	843-898	10-45
copper	843-898	12-45
Steel, C' steel	1010-1148	8-45
Stainless steels	1033-1287	30-60
Fe (ferrit)	1204-1482	10-600
nickel	1010-1148	30-45
Alnico magnets	1204-1301	120-150
Tungsten Carbides	1426-1482	20-30
Molybdenum	2054	120
Tungsten	2343	480

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[6] TS EN ISO 3923-1 Metal tozlar- Görünür yoğunluk tayini- Bölüm 1: Huni metodu (Metallic powders
- Determination of apparent density - Part 1: Funnel method)

[7] TS EN ISO 4490 Metalik tozlar-Kalibre edilmiş huni vasıtası ile akış süresinin tayini (hall flowmetre) ölçme metotları (Metallic powders - Determination of flow time by means of a calibrated funnel (Hall flowmeter))